Printable Datasheet

BASE LAMINATES USED	
POLYMIDE/BT*	Woven glass fibre
FR4ML*	Woven glass fibre / Epoxy
FR4 Rigid*	Woven glass fibre / Epoxy
FR2*	Paper / phenolic
CEM3*	Composite / random glass fibre
CEM1*	Composite / epoxy paper

(*All laminates are available in a range of thickness and copper foil weights)

BOARD AND PANEL SIZES	Metric	Imperial
Maximum panel dimensions	610 x 355 mm.	24 x 14 in.
Minimum panel dimensions	150 x 150 mm.	6 x 6 in.
Maximum board dimensions	560 x 305 mm.	22 x 12 in.
Minimum board dimensions	13 x 6 mm.	0.5 x 0.25 in.

HOLE DIAMETERS TOLERANCE RANGE	Metric	Imperial

Drilled Only:

Up to 1 mm. (0.039 in.)	0.025 mm.	0.001 in.
1 mm. to 2mm. (0.039 in./0.078 in.)	0.038 mm.	0.0015 in.
Above 2 mm. (0.078 in.)	0.050 mm.	0.002 in.
Plated Holes:		
Up to 1 mm. (0.039 in.)	0.050 mm.	0.002 in.
1 mm. to 2mm. (0.039 in./0.078 in.)	0.076 mm.	0.003 in.
Above 2 mm. (0.078 in.)	0.087 mm.	0.0035 in.

DRILLED HOLE POSITION TOLERANCE		
Hole positions located from film	0.076 mm.	0.003 in.

MACHINE CUTTING TOLERANCE		
NC Routing	0.1 mm.	0.004 in.

REGISTRATION TOLERANCE (Pad/Hole Ratio)		
Drill positional tolerance	0.076 mm.	0.003 in.
Pad positional tolerance (photo)	0.025 mm.	0.001 in.
Photo location tolerance	0.076 mm.	0.003 in.
Etch factor	0.0013 mm.	0.0005 in.
Minimum annular ring requirement	0.076 mm.	0.003 in.
TOTAL MANUFACTURE TOLERANCE	0.254 mm.	0.0105 in.

(This does not take into account mis-registration within artworks)

CAPABILITIES AND PREFERENCES

	Preferred		Minimum	
SPECIFICATION	mm.	inches.	mm.	inches
Track Width Gap	0.2	0.008	0.15	0.006
Minimum land around hole	0.2	0.008	0.15	0.006
Annular Ring Clearance	0.3	0.012	0.2	0.008
Minimum hole diameter	0.4	0.016	0.3	0.012
Minimum clearance on Solder resist			0.08	0.003
Minimum line width on legend			0.02	0.008
	Corner markers on external copper layer. Inside edge designates board profile. or			
Line around profile. Centre designates board profile.	0.15	0.006		

ADDITIONAL USEFUL INFORMATION

Text file showing which file refs. relate to which layer.

Format for Gerber and drill files.

NOTATION COMPONENT IDENTS	Metric	Imperial

(Legends available in White, Yellow and Black)

Alignment postional tolerance	0.2 mm.	0.008 in.
Minimum letter width on photoplot		
In copper	0.25 mm.	0.010 in.
Screen printed	0.2 mm.	0.008 in.

FINISHES AVAILABLE

- Copper
- Solder levelling
- Electroless Gold on Nickel
- Sterling Silver
- Fused tin lead
- Gold 2.5 microns minimum
- Immersion Tin

SOLDER MASK FINISHES AVAILABLE

Colours: Green, Blue, Black, Red or White

- UV Cured
- 2 part Epoxy
- Photo-imageable

Minimum annular ring	Metric	Imperial

Screen printed	0.25 mm.	0.010 in.
Photo-imaged	0.076 mm.	0.003 in.

SPECIAL SERVICES

- Any job which requires a finish or process which is non-standard.
- Help in the layout of the PCB to obtain the most economical method of manufacture.
- Help and advice on Multilayer layout and build.

COMMUNICATIONS

- Modem for CAD files (0)1223 425201

Set up requirements are as follows:

Baud Rate: 14400	Duplex: Full	Bits: 8
Parity: None	Stop Bit: 1	Emulation: ANSI
Password: hello		

- Email address: sales@cambridge-circuit.co.uk
- Telephone: 01223 423100
- Fax: 01223 425001

PHOTOPLOTTING / PHOTOGRAPHY

Full photoplotting and photography facility (including step and repeat)

- Barco Cresent 30 Laser Imager
- IGP Lazerline 25-S processor
- Littlejohn Type 49 Contact Printer
- Lavenir CAMmaster V7 Software

DRILL SHOP

- Wessel Pro-Sys 1 Automatic Load / Unload CNC Drill
- Micromat twin spindle auto-tool change CNC Drill / Router
- Micromat single spindle auto-tool change CNC Drill / Router
- Exellon Unidrill
- Twin Spindle Tooling Hole Drill
- Wessel Auto de-pinner
- JWA Quick - Score scoring machine
- Fag precision saw
- 1.3 metre P & S Guillotine

PHOTOMECHANICAL

- One 12" Hot Roll laminator
- Two 18" Hot Roll laminators
- Cirgraphics T3000 exposure unit
- Cirgraphics T4000 exposure unit
- Two GSPK Conveyorised developers
- Somaca double sided board cleaner
- Folex Helamat Diazo Developer

MULTILAYER

- Mackey Bowley Hydraulic Press
- Moore Cooling Press
- Inner Layer treatment and De-smear line

PLATING

- GSPK Shadow Direct metallisation line
- Manual copper / tin-lead plating line
- Acid gold plating line
- Ammoniacal etch line
- GSPK Dry film stripper with filter system
- FSL 20-24E Tin Lead Strip line

- Hibass Wash / Dry Unit
- Lantronic Hot Air Solder Levelling machine
- Post HASL cleaning line

SCREEN PRINTING

- Precision manual screen printers
- Conveyorised Ultra Violet curing system
- Thermal cure ovens

INSPECTION / LABORATORY

- Emma EL6146E Fixtureless Bare Board Tester
- Barco AccuMatch AOI
- Vision Engineering TS4 Stereo Microscope
- Coppertron SM6000 Copper thickness tester
- Comprehensive testing and inspection facilities conforming to **BS EN ISO 9002**

QUOTATION NOTES

All quotations are confirmed by Fax, Email or given over the Telephone if required.

For your quotations we require the following information:

1. Part number and issue level.
2. Quantity required.
3. Delivery lead time.
4. Dimensions of PCB.
5. Number of layers.
6. Approximate number of holes, number of different hole sizes and the smallest hole size.
7. Minimum track width and gap.
8. Type of material, board thickness / build and copper weights.
9. Area of gold plating if required.
10. Hot air solder levelling if required.
11. Solder mask type if required, on one or both sides.
12. Silkscreen (component ident) if required, on one or both sides.
13. Details of any slots, cut outs or other machining.
14. Type and form of data supplied e.g. Gerber or 1:1 photography.
15. Number of surface mount pads on each side.
16. Bare Board Test, number of pads on each side. *
17. Certificate of Conformity required.

* All Multilayer boards are Bare Board Tested.

LEAD TIMES QUOTED DO NOT INCLUDE DELIVERY TIME.

Delivery would normally be by overnight carrier, or special delivery services can be arranged.

SINGLE SIDED, DOUBLE SIDED, PTH and MULTILAYER

5 DAYS DELIVERY	(STANDARD)
4 DAYS DELIVERY	(25% PRICE LOADING)
3 DAYS DELIVERY	(50% PRICE LOADING)
2 DAYS DELIVERY	(75% PRICE LOADING)
1 DAY DELIVERY	(100% PRICE LOADING)
8 HOUR DELIVERY	(125% PRICE LOADING)

10 DAYS DELIVERY	LARGER PRODUCTION
15 DAYS DELIVERY	QUANTITIES

- All delivery times stated are in working days.
- Delivery times include photoplotting and commence from receipt of accurate finalised data.
- Fast reliable delivery of quality PCBs is acheived because we are

flexible enough to meet customer requirements.

- All lead times quoted do not include transport time.